Manvansh Singh

Zirakpur, Punjab | 9896838285 | manvansh086@gmail.com
LinkedIn | GitHub | Portfolio

Education

Chandigarh Engineering College, CGC Landran — Landran, Punjab
B.Tech in Robotics & Artificial Intelligence — 07/2024 – Present

Technical Skills

Languages: Python, C/C++, JavaScript
ML & Data: Regression, Classification (Random Forest, XGBoost), Anomaly Detection (Isolation Forest), Feature Engineering, Hyperparameter Tuning (GridSearchCV)
ML/DL: PyTorch, CNNs, U-Net, Computer Vision, Medical Image Segmentation
Frontend: HTML, CSS, React.js, Tailwind CSS
Backend & Database: Node.js, Express.js, FastAPI, MongoDB, MySQL, PostgreSQL
Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-learn, NiBabel, Plotly
Developer Tools: Git, GitHub, Postman, VS Code, Jupyter Lab, Vercel, Render

Experience

Devantra Innovations Pvt Ltd. — Mohali
Full Stack Developer — Jan 2026 – Present

- Built a full-stack Task Management module for an Enterprise ERP system, adopted by 20+ employees and 5 admins – architected 11 REST API endpoints using Node.js & PostgreSQL with role-based access for Admin, Manager, and Member roles.
- Built a Change Request workflow module end-to-end with 7 API endpoints handling task deadline extension requests, approvals, and rejections.
- Built a Change Request workflow module end-to-end with 4 API endpoints handling task deadline extension requests, approvals, and rejections.
- Built reusable, responsive UI components integrated across 7+ ERP pages with full frontend-backend wiring.

Devantra Innovations Pvt Ltd. — Mohali
MERN Stack Developer Intern — Aug 2025 – Jan 2026

- Built and styled 5+ reusable admin panel components, focusing on layout, usability, and responsive design.
- Implemented backend functionality using Node.js and Express.js to support admin page features.
- Collaborated within a 4-member development team using Git/GitHub for version control and feature integration.

Projects

Turbofan Predictive Maintenance System | *Python, FastAPI, XGBoost, Scikit-learn, React* — GitHub

- Engineered 46 time-series features (rolling stats, RUL labels) from NASA's C-MAPSS dataset (100 engines, 20K+ cycles) with engine-level splitting to prevent temporal data leakage
- Trained RF/XGBoost models (GridSearchCV) achieving R^2=0.79 (RUL), F1=0.87, Recall=0.93; Isolation Forest flagged failures 12 cycles out vs. 113 for normal readings

Dental CBCT Tooth Segmentation Pipeline | *PyTorch, U-Net, Medical Imaging* — GitHub

- Built a full preprocessing-to-postprocessing 2D U-Net pipeline (7.7M params) for 3D CBCT tooth segmentation, trained on 14,790 slices from 50 volumes on RTX 4050 (CUDA)
- Achieved 0.7479 Dice / 0.6948 IoU on held-out test set with custom PyTorch DataLoader and interactive 3D Plotly visualization of predictions

AI Interview Prep Platform | *MERN, JWT Auth, Gemini API* — GitHub

- Built an AI platform generating 8-10 role/experience-tailored interview questions via Gemini API, with "load more" (+4-5) and a pin/bookmark revision feature
- Enabled JWT authentication with protected routes and session expiry; deployed on Vercel and Render

Achievements

100+ LeetCode Problems Solved – consistent DSA practice across arrays, strings, hashing, and binary search
Finalist – Smart India Hackathon (SIH), Internal Round, 2025
1st Prize – CodeQuest (College-level Coding Competition)